



# Bottleneck identification for constraint relaxation in resource-constrained project scheduling

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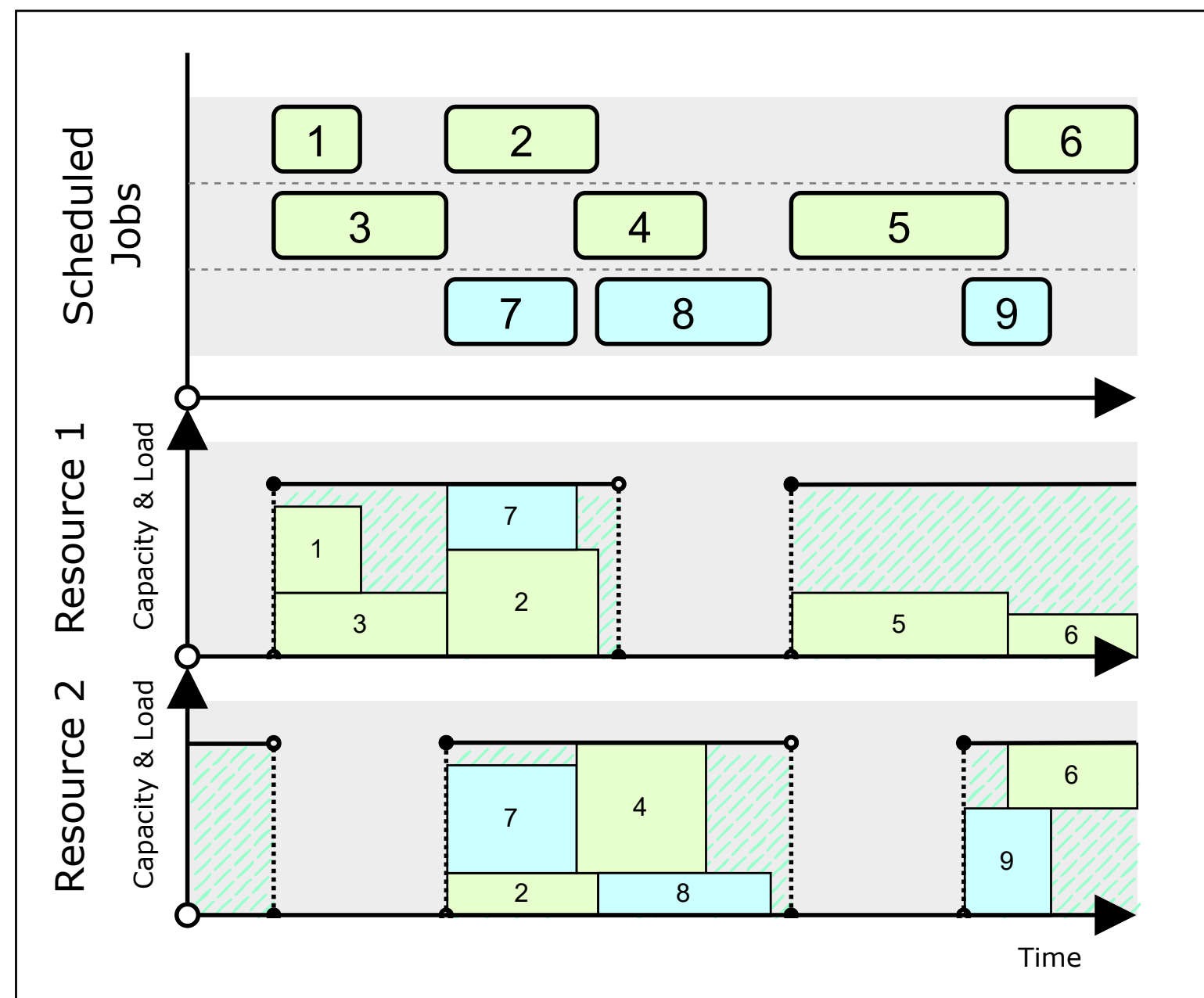


## 1. Problem statement

Scheduling large manufacturing systems is a difficult task. A commonly occurring problem is reducing the tardiness of a selected manufacturing order.

To address this problem, we model a manufacturing system as the Resource-Constrained Project Scheduling Problem (RCPSP), consisting of *jobs* with associated *precedences*, where jobs are executed on several *resources*. The problem is extended with time-variable resource capacities for modeling working shifts.

We use constraint programming for obtaining solutions to models of the problem. Resource constraints of those models are the focus of relaxations in this thesis.



Example of extended RCPSP schedule

## 2. Goals

- Design a method for identifying bottlenecks in obtained schedules.
- Propose model constraint relaxations related to the bottleneck resources in the obtained schedules. Such relaxations should improve the tardiness of a specific target manufacturing order while not modifying the original schedule much. Additionally, the relaxations should correspond to feasible and practical real-world implementations.

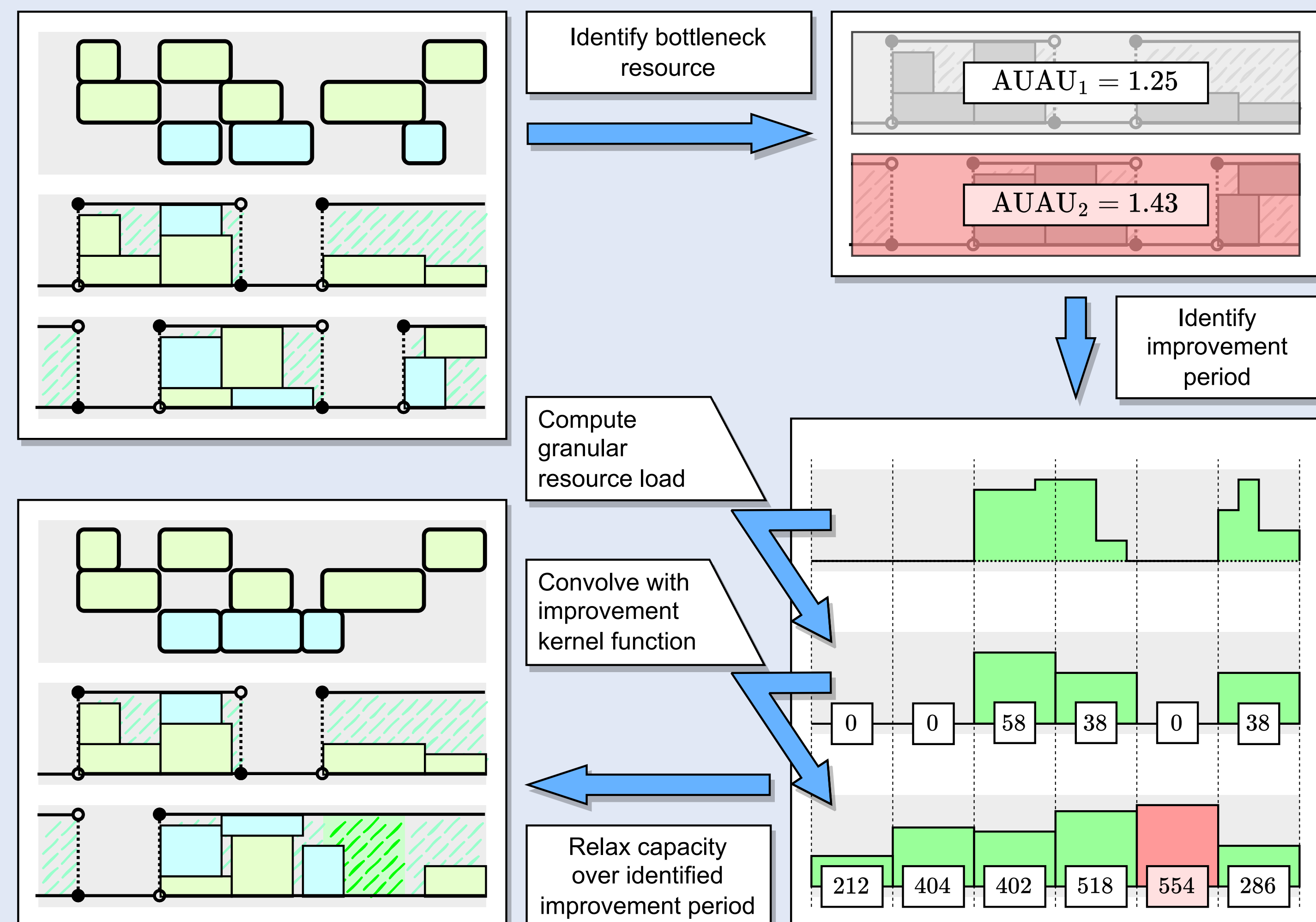
## 3. Identification Indicator-based Relaxing Algorithm

We propose two *bottleneck identification indicators* for the extended RCPSP — Machine Resource Utilization Rate (MRUR) and Average Uninterrupted Active Utilization (AUAU). Each computes a single numeric value for a given resource  $k$ .

$$\text{MRUR}_k = \frac{\text{total consumption of } k \text{ by all jobs}}{\text{total capacity of } k \text{ over project makespan}}$$

$$\text{AUAU}_k = \text{average utilization over uninterrupted active periods}$$

The Identification Indicator-based Relaxing Algorithm (IIRA) works as follows:



The proposed resource capacity functions are reduced to only include utilized changes. The changes are then divided into *migrations* and *additions* based on available capacities on remaining resources.

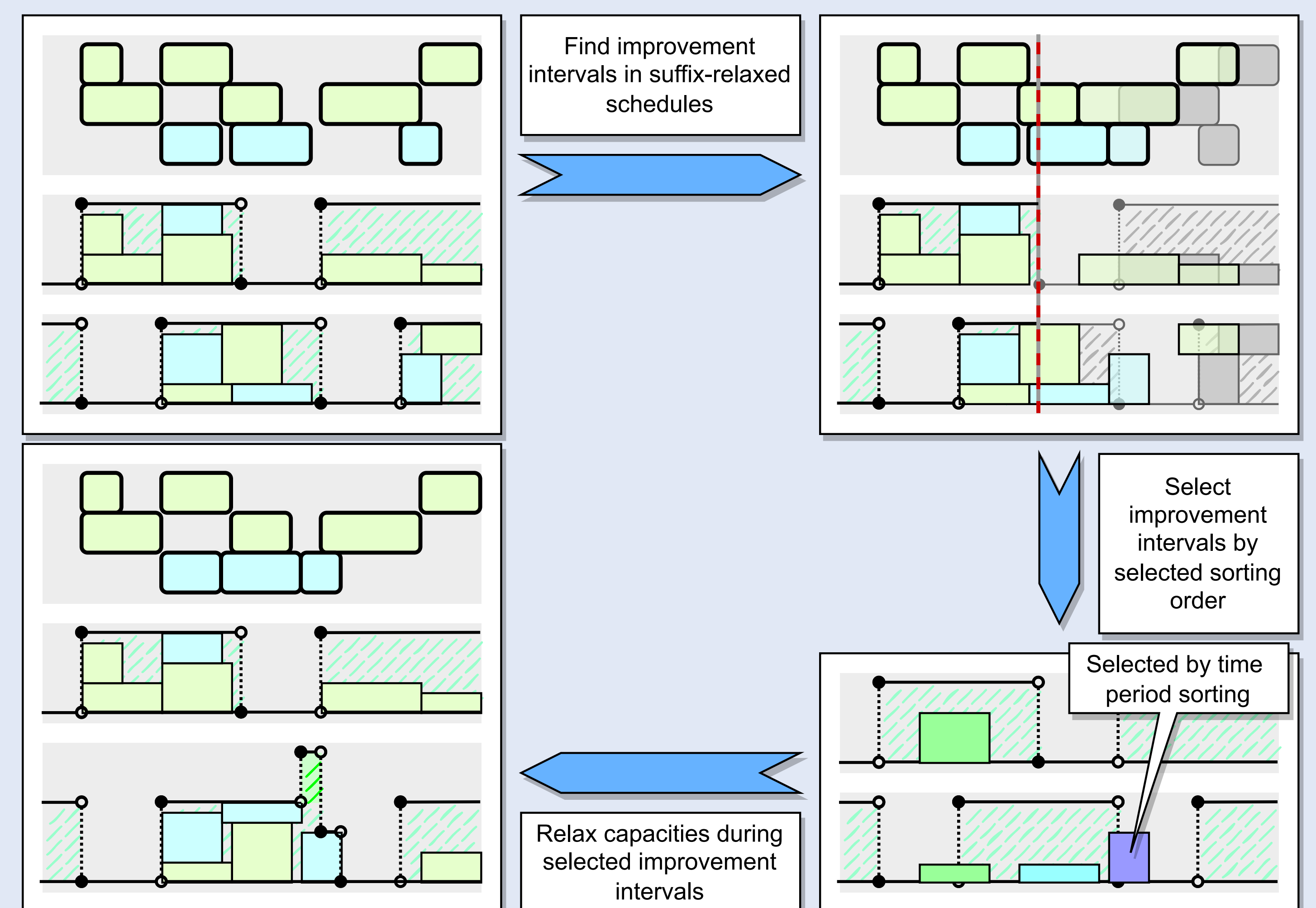
## 5. Experiments

Numerical experiments were conducted to analyze the capabilities of the proposed methods. For that purpose, we designed a set of problem instances (based on PSPLIB instances) modeling the used extension of the RCPSP.

We observed that the **SSIRA is more consistent in finding improving solutions** than the IIRA, while the **IIRA is able to find great improvements with low modification costs** on instances, where the SSIRA finds the same improvements with greater costs.

## 4. Schedule Suffix Interval Relaxing Algorithm

The Schedule Suffix Interval Relaxing Algorithm (SSIRA) relaxes *suffixes of schedules* to obtain heuristic interval relaxation proposals. The algorithm focuses on the target manufacturing order by identifying improvement intervals only within its left-shift closure. The *left-shift closure* of a manufacturing order is the set of jobs that, corresponding to various definitions, prevent the order from finishing earlier.



Same as in the IIRA, the capacity changes are reduced and corresponding capacity *migrations* and *additions* are identified.

## 6. Summary

In this thesis, we

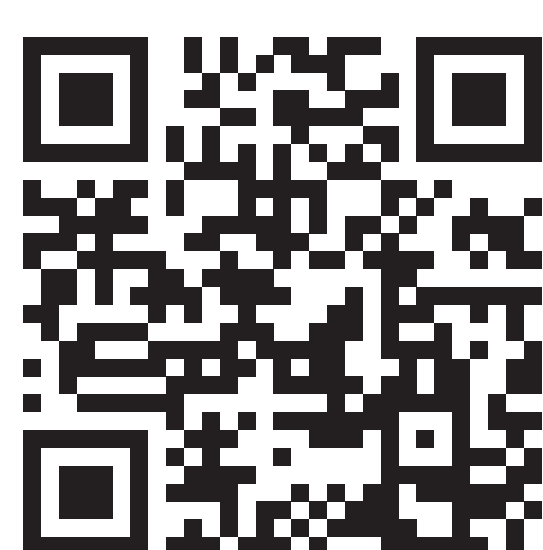
- **designed the IIRA** as a baseline solution adapting existing bottleneck identification approaches from the scheduling literature and proposing relaxations based on those identifications,
- **designed the SSIRA** as a novel approach to identifying schedule bottlenecks and relaxing related constraints in the RCPSP, and
- **conducted numerical experiments** showing that the IIRA achieves great improvements for small costs, while the SSIRA is more reliable in achieving improvements across various problem instances.

Future work could focus on modeling the relaxations as an optimization problem to obtain exact proposals capturing all model dependencies.

## Supervisor

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## Project repository

<https://github.com/Krtiik/RCPSPSandbox>

<https://github.com/Krtiik/Bottleneck-identification-for-constraint-relaxation-in-resource-constrained-project-scheduling/blob/main/thesis.pdf>

## Bachelor thesis

